RELATIVE FINITE PRESENTATION IS CHECKED ON AFFINE OPENS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 16411-16450, immediately following the statement at 16395-16409.

Editorial note. The statement contains the author's phrasing 'for every affine opens ... with f(U) subset V' and 'for every open subschemes'; both are retained verbatim.

Editorial note. Four cross-chapter pointers (More on Algebra, Schemes) plus the chapter's own definition are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Definition of a relatively finitely presented sheaf. chapter definition definition-relatively-finitely-presented-sheaf, cited at proof line 16411 onward. Fixes the property that the affine-local conditions must be shown equivalent to.
- Base change of relative finite presentation. More on Algebra, lemma more-algebra-lemma-base-change-relative-finite-presentation, cited in the proof. Moves the condition across the morphism to the affine level.
- Localising and glueing relative finite presentation. More on Algebra, lemmas more-algebra-lemma-localize-relative-finite-presentation and more-algebra-lemma-glue-relative-finite-presentation, cited in the proof. Provides the local-to-global step: local presentations glue along the affine cover.
- Standard open coverings of an affine scheme by two affines. Schemes, lemma schemes-lemma-standard-open-two-affines, cited in the proof. Supplies the two-affine cover used to verify the overlap conditions.

Source reference. Original definition 05H1: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16375–16375. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 065B: focused-seeds/geometry/S9-morph-rel-finite-presentation-characterize/ref-tags.txt, source lines 4–4. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 065D: focused-seeds/geometry/S9-morph-rel-finite-presentation-characterize/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 065A: focused-seeds/geometry/S9-morph-rel-finite-presentation-characterize/ref-tags.txt, source lines 6–6. Navigation pointer

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01IW: focused-seeds/geometry/S9-morph-rel-finite-presentation-characterize/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $\mathcal{F}$ be a quasi-coherent $\mathcal{O}_X$-module. The following are equivalent*

(1) *$\mathcal{F}$ is of finite presentation relative to S,*
(2) *for every affine opens $U \subset X$, $V \subset S$ with $f(U) \subset V$ the $\mathcal{O}_X(U)$-module $\mathcal{F}(U)$ is finitely presented relative to $\mathcal{O}_S(V)$.*

Moreover, if this is true, then for every open subschemes $U \subset X$ and $V \subset S$ with $f(U) \subset V$ the restriction $\mathcal{F}|_U$ is of finite presentation relative to V.

Proof. The final statement is clear from the equivalence of (1) and (2). It is also clear that (2) implies (1). Assume (1) holds. Let $S = \bigcup V_i$ and $f^{-1}(V_i) = \bigcup U_{ij}$ be affine open coverings as in Definition 05H1. Let $U \subset X$ and $V \subset S$ be as in (2). By More on Algebra, Lemma 065D it suffices to find a standard open covering $U = \bigcup U_k$ of U such that $\mathcal{F}(U_k)$ is finitely presented relative to $\mathcal{O}_S(V)$. In other words, for every $u \in U$ it suffices to find a standard affine open $u \in U' \subset U$ such that $\mathcal{F}(U')$ is finitely presented relative to $\mathcal{O}_S(V)$. Pick i such that $f(u) \in V_i$ and then pick j such that $u \in U_{ij}$. By Schemes, Lemma 01IW we can find $v \in V' \subset V \cap V_i$ which is standard affine open in V' and V_i. Then $f^{-1}V' \cap U$, resp. $f^{-1}V' \cap U_{ij}$ are standard affine opens of U, resp. U_{ij}. Applying the lemma again we can find $u \in U' \subset f^{-1}V' \cap U \cap U_{ij}$ which is standard affine open in both $f^{-1}V' \cap U$ and $f^{-1}V' \cap U_{ij}$. Thus U' is also a standard affine open of U and U_{ij}. By More on Algebra, Lemma 065A the assumption that $\mathcal{F}(U_{ij})$ is finitely presented relative to $\mathcal{O}_S(V_i)$ implies that $\mathcal{F}(U')$ is finitely presented relative to $\mathcal{O}_S(V_i)$. Since $\mathcal{O}_X(U') = \mathcal{O}_X(U') \otimes_{\mathcal{O}_S(V_i)} \mathcal{O}_S(V')$ we see from More on Algebra, Lemma 065B that $\mathcal{F}(U')$ is finitely presented relative to $\mathcal{O}_S(V')$. Applying More on Algebra, Lemma 065A again we conclude that $\mathcal{F}(U')$ is finitely presented relative to $\mathcal{O}_S(V)$. This finishes the proof. □